UIT University

BACHELOR OF SCIENCE (COMPUTER SCIENCE / SOFTWARE ENGINEERING)

CSC-204 DATABASE SYSTEMS

PROJECT REPORT

TITLE:

INDUSTRIAL PROCESS MANAGEMENT DATABASE

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PROJECT TITLE: INDUSTRIAL PROCESS MANAGEMENT DATABASE

- ## ABSTRACT:

The Industrial Process Management Database was developed to solve a common but critical problem in manufacturing: the disconnect between procurement, factory operations, and final delivery. In many traditional industrial settings, data is siloed across separate paper logs or independent spreadsheets, making it nearly impossible for leadership to track a project's true status or calculate accurate costs in real-time.

Our solution is a centralized Database Management System (DBMS) built on a normalized SQLite architecture. The system acts as a "Single Source of Truth" by pulling every stage of the manufacturing lifecycle into one cohesive digital thread. We replaced fragmented records with a unified workflow where every action is dependent on the completion of the previous one.

Key innovations in this system include:

- **Dynamic Asset Auditing:** A live Machine Manager that allows directors to register new factory equipment on the fly and track granular usage logs, showing exactly which worker utilized which machine and for how long.
- **Unified Logistics Pipeline:** A redesigned shipping module that merges packing and delivery into a single sequential process. This ensures that no product can be delivered without first being verified and packed by assigned personnel.
- **Automated Project Closure:** A logic-driven "Definition of Done" where the successful delivery of a product automatically triggers the completion of the entire project proposal and updates the financial status.
- **Director's Control Hub:** A multi-tier secure interface giving directors total visibility over vendor payments, workforce task allocation, and client feedback loops.

By enforcing strict relational constraints and eliminating over eight redundant data tables, we have created a high-performance system that prevents operational errors before they happen. This project demonstrates how a well-structured database can transform a "scratch-to-product" workflow from a series of messy, disconnected stages into a transparent, data-driven industrial pipeline.

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1 INTRODUCTION:

1.1 The Challenge

In many industrial settings, data is scattered across different logbooks or spreadsheets. The office knows about the bills, the floor manager knows about the machines, and the warehouse knows about shipping, but they rarely talk to each other. This "information gap" makes it nearly impossible for a Director to see the big picture. Our system bridges these gaps, making sure that every stakeholder is part of one single, transparent workflow.

1.2 Our Solution

We chose a relational database approach using SQLite to build a "Single Source of Truth." We didn't just want to store data; we wanted to enforce the rules of the factory. Our system uses strict constraints so that a project cannot move to "Delivery" if it hasn't been "Packed," and a task cannot be started if a "Worker" hasn't been assigned. This creates a foolproof digital thread for every project.

2 FEATURES:

2.1 Smarter Procurement & Finance

We moved away from simple inventory lists to a more comprehensive vendor tracking system. Every bill is logged with a status of 'Paid' or 'Not Paid,' allowing the Director to manage supplier relationships and project costs without leaving the dashboard.

2.2 Dynamic Machine Management & Auditing

One of our biggest updates is the "Live Machine Manager." Factory equipment is rarely static, so we built a way for the Director to add new machines (like CNC millers or 3D printers) on the fly. More importantly, we implemented an auditing system where every machine use is logged, showing exactly which worker used which tool and for how many hours.

2.3 The Unified Logistics Pipeline

We completely redesigned the shipping process. Instead of having separate tables for packing and delivery, we merged them into a single, sequential pipeline.

- **Step-by-Step Validation:** You cannot mark an item as delivered until the system verifies it was packed by an assigned staff member.
- **Automatic Closure:** Once the "Delivered" button is clicked, the system automatically marks the entire Project Proposal as 'Completed,' closing the loop on that project.

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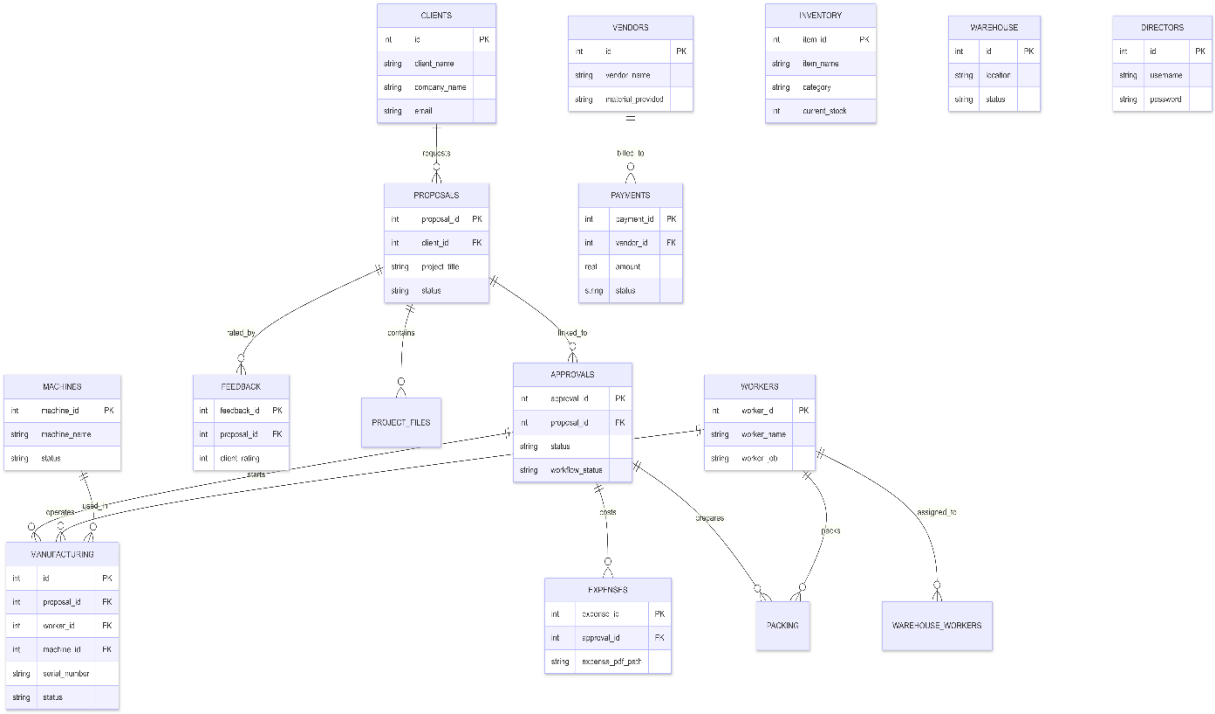
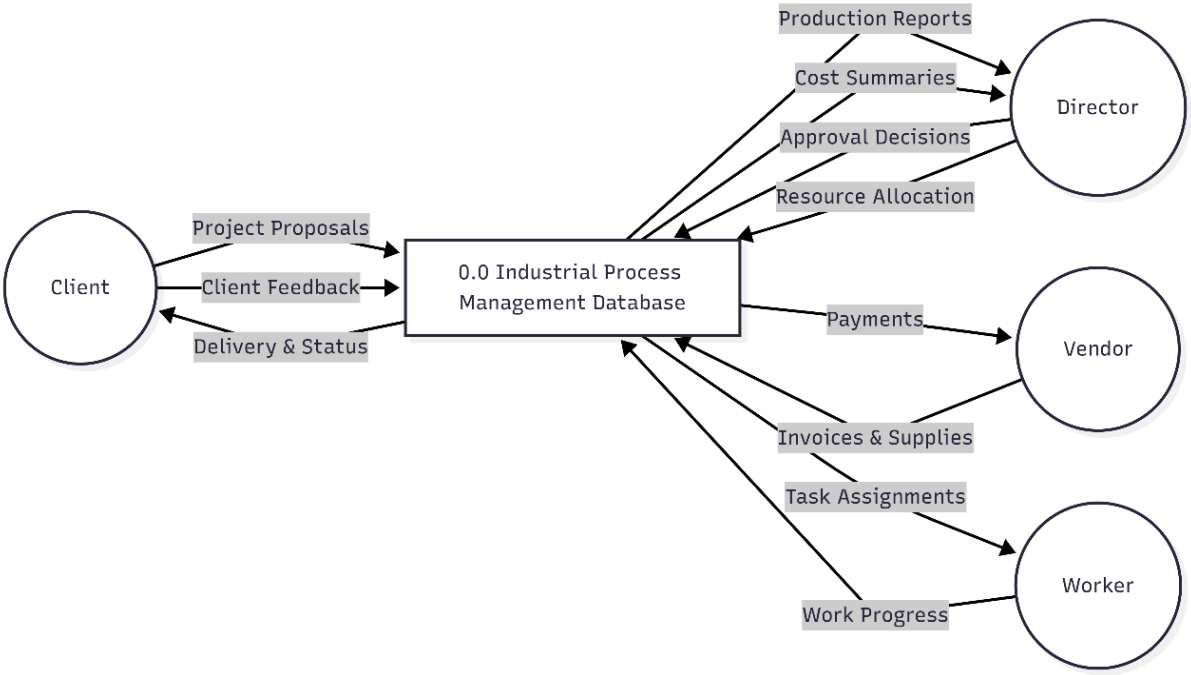
3 DATABASE DESIGN

3.1 ERD DIAGRAM

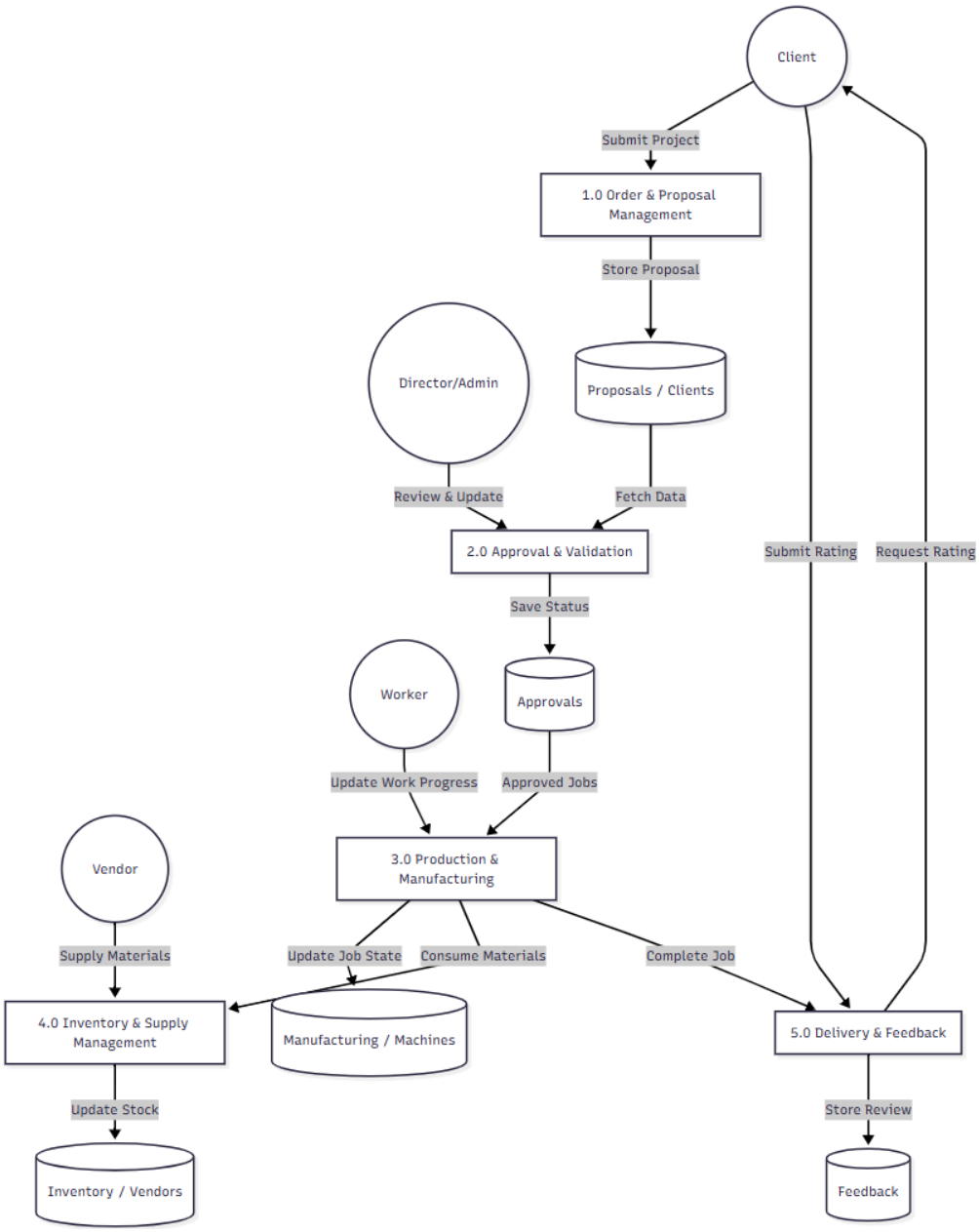


4 DATA FLOW DIAGRAM (DFD)

4.1 CONTEXT DIAGRAM



4.2 LEVEL 1 DFD



5 IMPLETATION:

5.1 Schema Normalization

We cleaned up our database significantly, removing over 8 redundant tables that were cluttering the system. The new schema follows Third Normal Form (3NF) to prevent data duplication. The core of the database is the proposals table, which connects directly to the logistics, management (manufacturing), and approvals tables.

5.2 Security & Multi-Tier Access

We built distinct portals for different users:

- **The Director:** Has full control over approvals, hiring workers, adding machines, and viewing financial summaries.
- **The Worker:** Has a focused view where they can log machine time and update task statuses, but they are restricted from seeing sensitive financial or client data.

6 CONCLUSION

Developing this system has shown us that even the most complex industrial cycles can be tamed with a well-designed database. By centralizing the data, we've given the Director total transparency—knowing exactly where a project is at any given moment. This isn't just a digital version of paper logs; it's a proactive management tool that prevents errors before they happen.

Looking forward, this database is perfectly structured to handle future upgrades, like automated machine sensors or predictive maintenance alerts based on the usage logs we are already collecting.

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